MIT | Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	COURSE SYLLABUS	
Department of Computer Science and Engineering(AIML)	**W.E.F**	**AY:** 2025 - 2026 (Rev. 2023 (NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY CSE(AIML)	**COURSE NAME**	ARTIFICIAL INTELLIGENCE
	COURSE CODE	2311212T
	COURSE CREDITS	2
RELEASE DATE : 01/07/2025	**REVISION NO.**	0.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
2	NIL	15	20	40	NIL	NIL	75

PREREQUISITE KNOWLEDGE: Problem-Solving Skills, Discrete Structure, Linear Algebra

COURSE OBJECTIVES:

2311212T.CEO.1: To introduce core AI concepts with a focus on ethical decision-making.

2311212T.CEO.2: To develop problem-solving skills through search strategies and intelligent decision-making.

2311212T.CEO.3: To equip students with essential feature engineering techniques to enhance AI model performance, while recognizing ethical use of data and responsible preprocessing.

2311212T.CEO.4: To illustrate diverse applications of soft and evolutionary computing across various domains and real world scenarios.

COURSE OUTCOMES:

After successful completion of the course, students will be able to,

2311212T.CO.1: Apply fundamental AI concepts to solve real-world problems[L3].

2311212T.CO.2: Apply appropriate search strategies to effectively solve AI problem solving scenarios across various domains[L3].

2311212T.CO.3: Utilize feature engineering techniques to enhance model accuracy across data types with ethical data handling [L3].

2311212T.CO.4: Make use of fuzzy logic and evolutionary algorithms for solving real-world problems in optimization and decision-making [L3].

COURSE ABSTRACT:

The Artificial Intelligence course provides a comprehensive understanding of the theoretical foundations and practical applications of AI and Machine Learning technologies. It covers key topics such as intelligent agents, search strategies, knowledge representation, feature engineering, fuzzy logic systems, evolutionary computation, and swarm intelligence.

Students will explore real-world applications, including autonomous systems, GPS navigation, medical diagnosis, and portfolio optimization, preparing them to tackle challenges in optimization, decision-making, and uncertainty management. Self-study topics and further readings foster independent learning and deeper insights into emerging AI trends.

This course emphasizes an implementation-based learning approach, where students will apply theoretical concepts through hands-on problem-solving, case studies, and real-world scenarios. The internal assessment and mid-semester evaluation will be designed accordingly to assess both conceptual understanding and practical application, ensuring that students gain industry-relevant skills.

UNIT 1	INTRODUCTION TO ARTIFICIAL INTELLIGENCE	06 HOURS

Applications/Case Study: Self-Driving Vehicles and Drones)

Contents: Foundation and history of Artificial Intelligence (AI), AI applications and types, Ethical and Social Impacts of AI, Intelligent Agents, Introduction to Reasoning and Knowledge Representation.

Self Study: Neural Network

Further Readings: Artificial General Intelligence.

UNIT 2	PROBLEM SOLVING THROUGH SEARCH STRATEGIES	10 HOURS

Applications/Case Study: GPS Navigation System, Chess Game

Contents: Search Strategies, State Space Search, Uninformed Search(Breadth-First, Depth-First), Heuristic Search (Greedy Best First Search, A* Algorithm, Admissibility of A*), Constraint Satisfaction Problem (CSP), Local Search, Minimax Search, Alpha-Beta Pruning, Ethical design of AI systems in decision-making, Transparency in pathfinding and decision logic, Case Studies and Applications: Deep Blue, AlphaZero, Pathfinding in Games, Maze Solving,Network Routing,Puzzle Solving, Game AI,Traveling Salesman Problem.

Self Study: Adversarial Search

Further Readings: Game Theory

UNIT 3	FEATURE ENGINEERING FOR MACHINE LEARNING	09 HOURS

Applications/Case Study: Stock Price Prediction

Contents: Introduction to Feature Engineering, Types of Features, Ethical use of features, Responsible data preprocessing, Feature Transformation: Missing Value Imputation, Scaling (Min-Max,Standardization), Encoding (One-Hot, Label), Feature Selection: Chi2 square method, Variance Threshold, Recursive Feature Elimination, Feature Construction: Polynomial Features, Text and Image Feature Extraction, Time-Series Feature Engineering, API Integration for Time-Series Data.

Self Study: Principal Component Analysis (PCA)

Further Readings: Automated Feature Engineering and Ethical Data Practices

UNIT 4	SOFT AND EVOLUTIONARY COMPUTING	05 HOURS

Applications/Case Study: Fuzzy Logic in Washing Machines/ Autonomous Vehicles
Contents: Introduction to Soft Computing, Fuzzy Logic: Membership Functions, Fuzzy Rules and Reasoning, Applications of Fuzzy Logic, Evolutionary Computation: Swarm Intelligence, Ant Colony Optimization(ACO).
Self Study: Particle Swarm Optimization (PSO)
Further Readings: Genetic Algorithms and Bio-Inspired Ethical AI

TEXT BOOKS

1. Stuart Russel and Peter Norvig (2022). *Artificial Intelligence: A Modern Approach.* 4th Edition, Pearson. ISBN 978-9356063570.
2. Rudolf Kruse, Christian Borgelt, Christian Braune, Sanaz Mostaghim, Matthias Steinbrecher (2023). *Computational Intelligence: A Methodological Introduction.* 3rd Edition, Springer. ISBN 978-3030422295.

REFERENCE BOOKS

1. Dr. Nilakshi Jain (2019). *Artificial Intelligence: Making a System Intelligent.* 1st Edition, Wiley.ISBN 978-8126579945.
2. Timothy J. Ross (2020). *Fuzzy Sets and Logic with Engineering Applications.* 4th Edition, Wiley.ISBN 978-8126502608
3. Adam Slowik (2022). *Swarm Intelligence Algorithms.* 1st Edition, CRC Press.ISBN 978-0367496142.
4. Melanie Mitchell (2020). *Artificial Intelligence: A Guide for Thinking Humans.* 1st Edition, Pelican. ISBN 978-0241404836.
5. Alice Zheng (2018). *Feature Engineering for Machine Learning: Principles and Techniques for Data Scientists, Second Edition.* 1st Edition, O'Reilly.ISBN 978-9352137114.

E-RESOURCES

1. https://nptel.ac.in/courses/106102220
2. https://nptel.ac.in/courses/108104157
3. Pioneering research on the path to Artificial General Intelligence (AGI),https://openai.com/research/
4. https://openlearning.mit.edu/courses-programs/mit-opencourseware
5. IEEE Transactions on Artificial Intelligence,"https://ieeexplore.ieee.org".
6. https://da.inf.ethz.ch/teaching/2022/CIL/